Storm Damage Risk Analytics

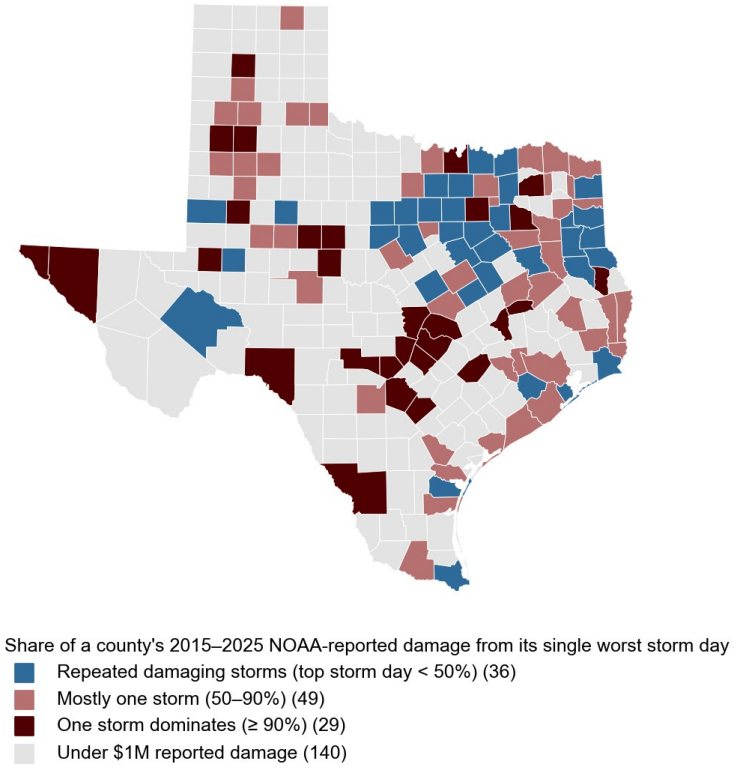
Visual analytics of recurring and catastrophic severe-weather damage across Texas counties, for insurance analysts

Team 8

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Preliminary mock-up insurviz.web.app

Code github.com/hendrosetyawan/insurviz

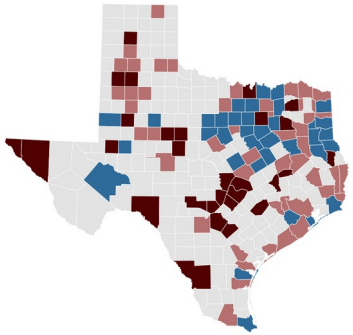


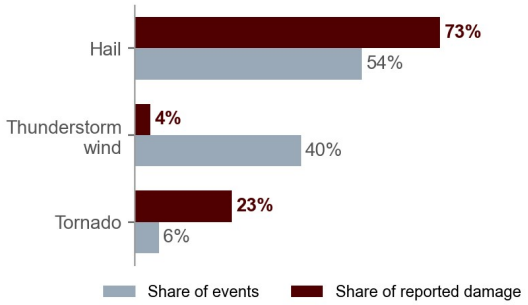
Share of each county's 2015–2025 NOAA-reported storm damage that came from its single worst storm day.

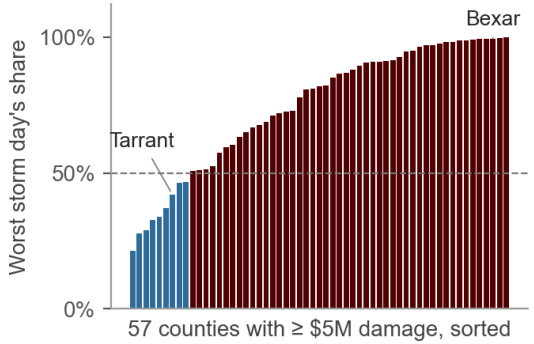
The Problem: Damage Totals Hide How Storm Losses Happen

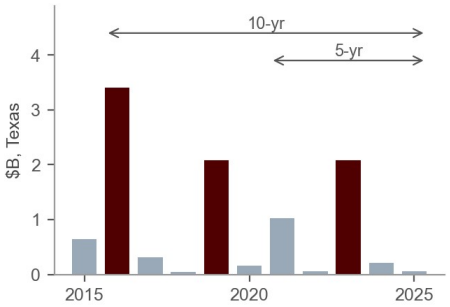
Domain Property insurance in Texas: historical damage from hail, thunderstorm wind and tornadoes.

Problem Underwriters and risk analysts must explain why storm losses differ across counties, but total damage and storm counts do not separate persistent exposure from rare catastrophic events.









Where

114 of 254 counties reported ≥ $1M damage, 2015–2025

Which hazard

Tornado: 6% of events, 23% of damage

Recurring or rare?

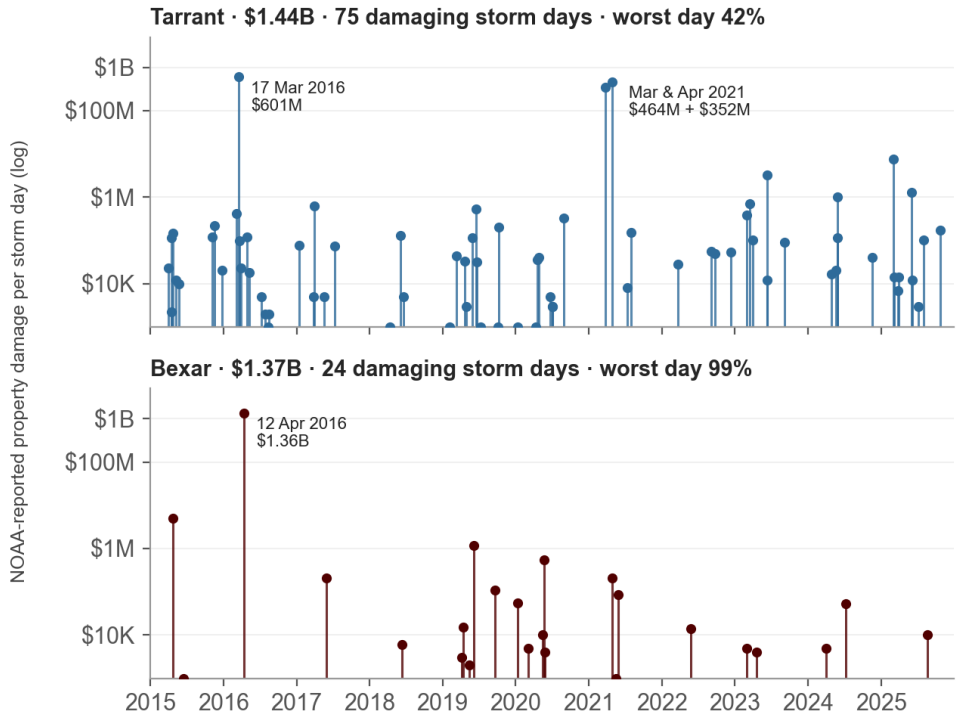
48 of 57 counties: one storm day ≥ half of damage

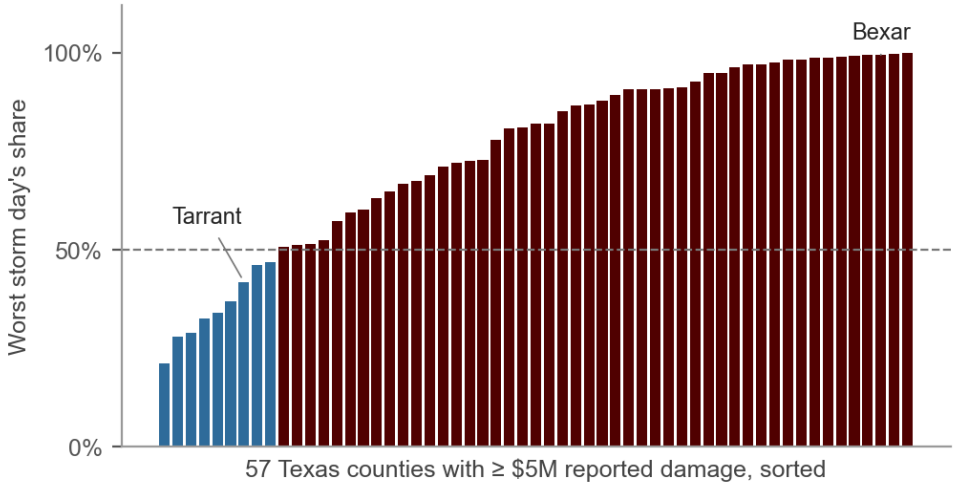
When

2016, 2019 and 2023 = 75% of 11 years' damage

Our proposal An interactive visualization that helps analysts compare counties, find the hazards behind historical damage, and tell recurring storm impacts from rare extreme events.

Why It Matters: Same Damage Total, Different Risk





$1.44B vs $1.37B

Tarrant vs Bexar, 2015–2025: three storms of $350M+ in two years vs one hailstorm (99%)

48 of 57

counties with ≥ $5M damage where one storm day caused at least half of it

4 in 5

Texas storm events with blank (15%) or $0 (64%) damage: reported damage ≠ insured loss

Target Users: Insurance Analysts and Homeowners

INSURANCE ANALYSTS (PRIMARY)

Catastrophe-Risk Analysts

Is a county's damage history driven by recurring storms or by a few extreme events?

Portfolio-Risk Analysts

Why does County A have more historical damage than County B under the same filters?

Underwriting Analysts

Which hazards and storms sit behind a location's loss history, and can I trust the record?

HOMEOWNERS AND COMMUNITIES (SECONDARY)

Homeowners and Home Buyers

Does my county see damaging storms every year, or was one storm the outlier?

Roof and Mitigation Decisions

Does recurring local hail justify an impact-resistant roof? TDI encourages insurers to discount such mitigation.

Community Groups and Local Officials

Which hazards and storms drive local damage, and is the record complete enough to plan on?

Why now Texas homeowners premiums rose 70% (median county, 2019–2025), and catastrophe models are calibrated on a short storm record that under-reports rural storms (Moody's, 2025). Analysts need to see the observed history behind their numbers.

Today's Insurance Tools: What They Leave Out

Xactimate

Verisk · claims estimating

Answers What will it cost to repair this damaged home? Line-item estimates from local price lists.

Where analysts struggle

- One claim at a time, after the loss
- No view of how often, or by which hazard, a county is hit
- Cannot compare counties' storm history

→ Motivates R1 · R3 · R5

360Value

Verisk · replacement cost

Answers What would this home cost to rebuild? Insurance-to-value at quote and renewal.

Where analysts struggle

- Values the building, not the hazard
- Storm history must come from elsewhere
- Raw damage totals ignore how built-up a county is

→ Motivates R1 · R6

Risk Modeler

Moody's RMS · catastrophe modeling

Answers What losses could a portfolio face? Simulated events → average annual loss and exceedance curves.

Where analysts struggle

- Modeled averages hide whether losses recur or come from the tail
- Hard to trace results back to real storms and years
- Specialist, licensed tool; vendor models can differ

→ Motivates R2 · R4 · R5 · R6

Common gap None of them shows the observed storm record behind a county's losses in one place. Our system complements these tools; it does not price repairs or model risk.

Sources: Verisk / Xactware product documentation; Moody's RMS (2025). Challenges are from public documentation and will be checked in the interviews.

User Interviews: Grounding the Requirements

WHO (PLANNED, n = 7)

Insurance practitioners (2)
Underwriter or claims adjuster at a Texas carrier or agency

Catastrophe or actuarial analyst (1)
Models storm losses for a portfolio

Public-sector reviewer (1)
TDI or Office of Public Insurance Counsel staff

Homeowners and community (2)
Recently re-roofed or non-renewed homeowner; neighborhood association or roofer

Risk & insurance faculty / students (1)
Pilot interview to test the protocol

HOW (45 MIN, SEMI-STRUCTURED)

1. **Workflow** 10 min
   Data and tools used today (e.g., Xactimate, 360Value, Risk Modeler)

2. **Pain points** 10 min
   What those tools cannot show; explaining a loss spike to managers

3. **Think-aloud** 20 min
   Six tasks on a prototype, one per requirement

4. **Rank & revise** 5 min
   Which requirement matters most; what is missing

WHAT WE ASK

"How do you tell recurring storm damage from a one-off catastrophe?"

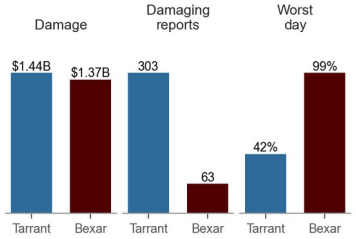
"What can't your current tools show you about a county's storm history?"

"Do storm records match what you see in claims?"

Homeowners: "Would knowing how often hail hits your county change your roof choice?"

Status Recruiting in October 2026; findings will confirm, rank or revise R1–R6. AI-simulated tool-user views helped draft questions only; they are not findings.

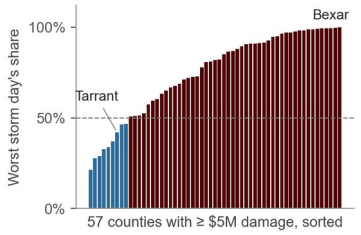
Design Requirements: Six Questions Insurance Analysts Ask



R1

Geographic loss comparison

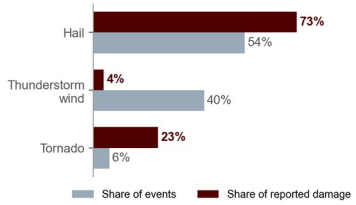
Why does County A have higher storm damage than County B?



R2

Recurring vs. catastrophic loss

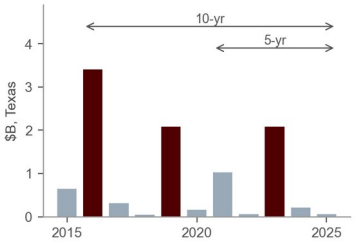
Regular damaging storms, or a few extreme events?



R3

Hazard contribution

Is the damage mainly hail, tornado or thunderstorm wind?



R4

Temporal loss pattern

Repeated over 5, 10, 20 years, or concentrated in one or two?

Date	County	Hazard	Damage
20 Oct 2019	Dallas	Tornado	$1.55B
12 Apr 2016	Bexar	Hail	$1.36B
20 Oct 2019	Dallas	Tornado	$400M
19 May 2023	Collin	Hail	$400M
08 May 2015	Hale	Hail	$300M

R5

Storm event investigation

Which storm events explain this county's high damage?



Damage field of 25,867 Texas events
(NOAA estimate, not insured loss)

R6

Data context and interpretation

Can I trust this comparison, and what are its limits?

Each requirement answers an analyst question and fills a gap in today's tools; the interviews will confirm, rank or revise them.

Data: Storm History, Insured Losses and Exposure

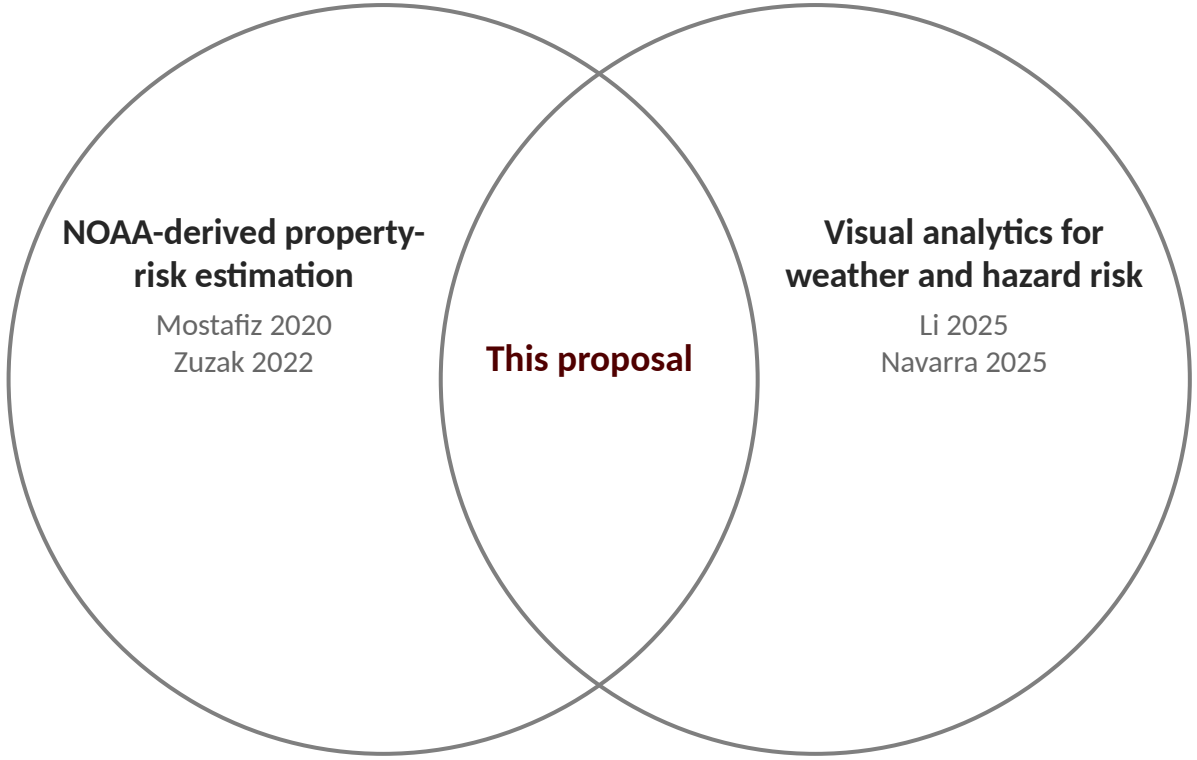
Dataset	Unit · years	Key fields	Role
NOAA NCEI Storm Events	Event → county · 2006–2025	Hail, thunderstorm wind, tornado; date, location, magnitude, damage estimate, narrative	Storm history (R1–R5)
TDI · Texas homeowners losses by county	County × year · 2019–2025	Paid losses by type (wind & hail, water, fire, other); policies in force	Insured-loss reference (R6)
U.S. Census ACS 5-year	County	Housing units, owner-occupied homes, median home value	Development context (R1, R6)
TDI · Premiums and HB 2067 reasons	County / ZIP · 2019–2026	Average premium; nonrenewal and declination reasons (e.g., roof, wind/hail)	Market context

Caution NOAA damage is an estimate, not insured loss; 15% of Texas events (2015–2025) leave it blank and 64% report $0. TDI losses cover homeowners policies only (no TWIA wind/hail). Dollars are nominal; we will add CPI-adjusted values.

Visualization Tasks & Encoding

R1	**Geographic loss comparison**	→	Compare counties side by side under shared filters
R2	**Recurring vs. catastrophic**	→	Summarize distribution; find outliers (ranked-event curve)
R3	**Hazard contribution**	→	Compare shares of events vs damage (paired bars)
R4	**Temporal loss pattern**	→	Identify trends and spikes (annual bars, 5/10/20-yr)
R5	**Storm event investigation**	→	Locate and look up events (linked table, map, narrative)
R6	**Data context**	→	Annotate reliability (missing data, NOAA vs TDI)

Gap & Potential Novelty



No tool we found shows whether a county's observed storm damage is recurring or driven by a few extreme events.

Potential Novelty One linked view of NOAA storm history for insurance analysts that separates recurring from catastrophic damage, ties it to hazards and storms, and flags data limits.

Mostafiz et al. (2020) Frontiers in Earth Science 8
Census-block property-loss risk from hail, tornado and more (Louisiana)

Zuzak et al. (2022) Natural Hazards 114(2)
FEMA National Risk Index: expected annual loss, 18 hazards, every county and tract

Li et al. (2025) IEEE TVCG 31(10)
Havior: visual analytics for heat-risk management (model data + news)

Navarra et al. (2025) Int. J. Disaster Risk Reduction 126
Visual analytics of social-media reports to detect extreme-weather impacts

Commercial insurance tools Xactimate · 360Value · Risk Modeler
Property-level costs or modeled losses; licensed; no view of observed storm history

Development Plan

MD
Data integration & context (R6)

NOAA 2006–2025, TDI and Census joins; data-quality flags

Aadi
County comparison (R1, R3)

Side-by-side county views; hazard breakdown

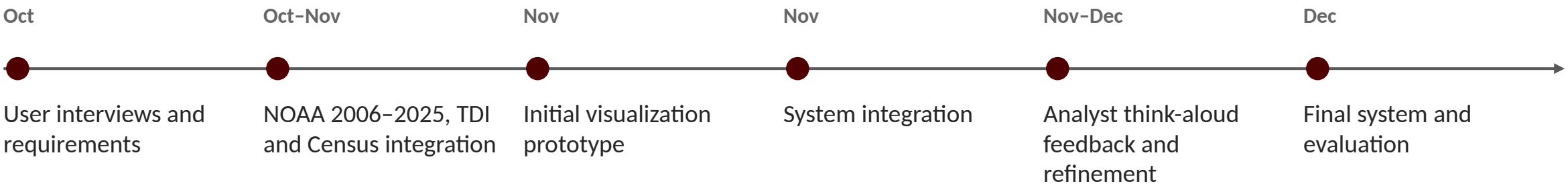
Hendro
Recurring vs. catastrophic (R2, R4)

Event-distribution and multi-year views

Abhinav
Event investigation (R5)

Event table, map links, view coordination

Shared tasks: user interviews (each member interviews at least one analyst), design decisions, evaluation, report, presentations.



Initial work: mock-up on NOAA, TDI and Census data insurviz.web.app • **Code** github.com/hendrosetyawan/insurviz

Reflection on Using AI

- **AI Usage**
  - Brainstorming visualization project domains
  - Refining the problem statement
  - Identifying NOAA as a possible reliable data source
  - Identifying related work search directions and organizing proposal content
  - Role-playing users of Xactimate, 360Value and Risk Modeler to draft interview questions (not used as findings)

Thank you! Any questions?